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Appendix A. Q-Superlinear Implies UP-Superlinear.

Proof of Theorem 3.2. Define

$$\xi_k = \|\mathbf{x}_k - \mathbf{x}_*\|, \quad f(k) = -\ln \xi_k.$$

By Q-superlinear convergence,

$$\lim_{k \rightarrow \infty} \frac{\xi_{k+1}}{\xi_k^q} = Q_q, \quad q > 1, \quad 0 < Q_q < \infty.$$

Taking logarithms, define

$$(A.1) \quad d(k) = f(k+1) - qf(k) = -\ln\left(\frac{\xi_{k+1}}{\xi_k^q}\right).$$

Then

$$\lim_{k \rightarrow \infty} d(k) = -\ln Q_q := d,$$

so that $\{d(k)\}$ is bounded; i.e., there exists $M > 0$ with $|d(k)| \leq M$.

Unrolling the recurrence

$$f(k+1) = qf(k) + d(k)$$

yields

$$f(k) = q^{k-1}f(1) + \sum_{j=1}^{k-1} q^{k-1-j}d(j).$$

Dividing by q^k , we have

$$\frac{f(k)}{q^k} = \frac{f(1)}{q} + \sum_{j=1}^{k-1} \frac{d(j)}{q^{j+1}}.$$

Since the tail of the series is bounded by

$$\sum_{j=k}^{\infty} \frac{|d(j)|}{q^{j+1}} \leq M \sum_{j=k}^{\infty} \frac{1}{q^{j+1}},$$

the limit

$$(A.2) \quad s = \lim_{k \rightarrow \infty} \frac{f(k)}{q^k} = \frac{f(1)}{q} + \sum_{j=1}^{\infty} \frac{d(j)}{q^{j+1}}$$